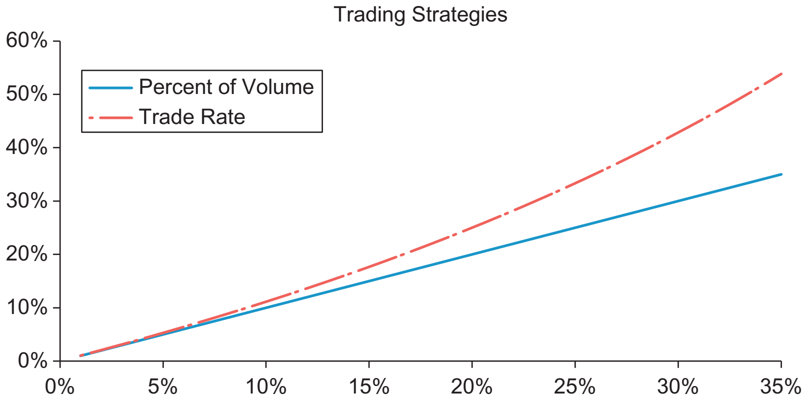

■ **Figure 7.1** Trading Strategies.

and represents the number of shares to trade in periods $1, 2, 3, \ldots, n$. The total number of shares executed over this period is $X = \sum x_i$. The advantage of the trade schedule is that it allows front-loading and/or back-loading of trades to take advantage of anticipated price movement, volume conditions, as well as effective risk management during a basket trade (these are further discussed in Chapter 9, Portfolio Algorithms).

Comparison of *POV* rate to Trade Rate

There is a direct relationship between the trading rate α and POV rate:

$$POV = \frac{\alpha}{1+\alpha} \text{ and } \alpha = \frac{POV}{1-POV}$$

A comparison of POV rate to α is shown in Figure 7.1. For POV less than 15% there is minimal difference in these two calculations. However, as we start increasing these rates, the measures start to deviate.

TRADING TIME

We define trading time in terms of volume time units. The value represents the percentage of a normal day's volume that would have traded at a given point in time. For example, if 1,000,000 shares trade on an average trading day, the volume time when 250,000 shares trade is: $t^* = 250{,}000/1{,}000{,}000 = 0.25$. The volume time when 1,250,000 shares trade is: $t^* = 1{,}250{,}000/1{,}000{,}000 = 1.25$.